

Reading and editing refusal in a model’s workspace

The model decides to refuse ten layers before it writes a word — and you can read it. But you can only partly erase it from the inside.

method jlens pullback ablation **baseline** activation-space **eval** AdvBench · XSTest · ARC
ablation

Abstract. We use the Jacobian lens to locate, read, and edit refusal in Qwen3.5-4B. At the generation position — before any output token — harmful prompts light up Cannot/无法/illegal in the workspace at layers 16–24 (refusal-mass $\approx +7$ vs. ≈ 0 for benign). We remove refusal by ablating the **lens pullback** of the refusal-token unembeddings, and compare it to activation-space ablation, measuring collateral **inside** the interpretable workspace. The pullback is far more surgical ($5.6\times$ less benign-workspace distortion) but removes less behavioral refusal: it clears the **verbalizable** “I-cannot” disposition while the model still refuses. Only $\sim\frac{1}{3}$ of behavioral refusal is workspace-mediated; full removal requires editing directions **outside** the workspace — exactly what ablation does, and what it pays for in collateral.

1 The idea

Activation-space *ablation* removes refusal by projecting out one direction — the mean difference between harmful and harmless activations — everywhere in the residual stream. That direction is derived from what correlates with harmful *input*, and its damage is only ever checked at the *output*.

The Jacobian lens gives a sharper handle. J_l is the average forward map from a layer- l residual to the final logits, so the residual direction that *causes future refusal* is the pullback of the refusal-token unembeddings:

$$d_l = J_l^\top (g \odot w), \quad w = \text{mean}(W[\text{refusal}]) - \text{mean}(W), \quad g = \text{final-norm gain}$$

This is refusal *as it lives in the verbalizable workspace*. Ablating d_l (via a forward hook, $h' = h - \alpha \cdot Q^\top(Qh)$, at every fitted layer ≥ 8 ; nothing written to weights) should remove the disposition to refuse while leaving the rest of the workspace untouched. We test two claims: that it is *more surgical* than ablation, and how much of refusal is workspace-mediated at all.

2 Refusal is legible before the first token

Reading the J-space at the generation position — before the model writes anything — harmful prompts light up Cannot / 无法 / illegal at layers 16–24; benign prompts do not. No harmful/harmless contrast set is needed to find it: the pullback of a handful of refusal tokens suffices.

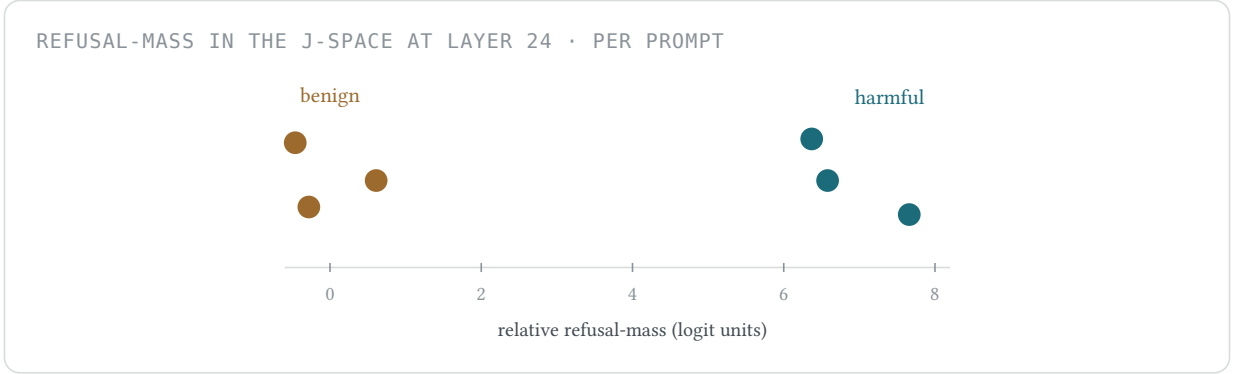


Figure 1: Each dot is one prompt. Harmful requests sit at +6 to +8; benign requests hover at 0 — the workspace has already “decided” to refuse ~10 layers upstream of any output.

3 Surgical, but partial

At edit strength 1 across all layers ≥ 8 , the pullback distorts the benign workspace **5.6× less** than ablation while suppressing workspace-refusal **2.2× more**. It is precisely the refusal-readout direction, so it leaves off-refusal content alone. The catch: it removes far less *behavioral* refusal.

edit	AdvBench ↓	XSTest-uns.	ARC	workspace KL ↓	refusal suppr. ↑
original	0.99	0.91	0.98	0.000	0.00
mean-diff (ablit- eration)	0.06	0.13	0.98	0.257	3.44
pullback	0.78	0.13	0.98	0.046	7.55
pullback sub- space r=3	0.55	0.23	0.98	0.196	7.18

Table 1: Strength-1 ablation on all layers ≥ 8 (n = 120 AdvBench / 200 XSTest / 250 ARC / 48 controls). **Workspace KL** is off-refusal-axis divergence of the J-space readout on benign controls — collateral measured inside the interpretable workspace. Every edit leaves ARC capability intact; the meaningful damage signal is the workspace.

A strength sweep makes the trade explicit. Neither method dominates: the pullback owns the low-collateral, partial-removal corner; ablation reaches full removal but drifts right into workspace collateral. Pushed to strength 3 both remove all refusal *and destroy the model* (ARC $\rightarrow$ 0.22), far off-chart at KL 6–17.

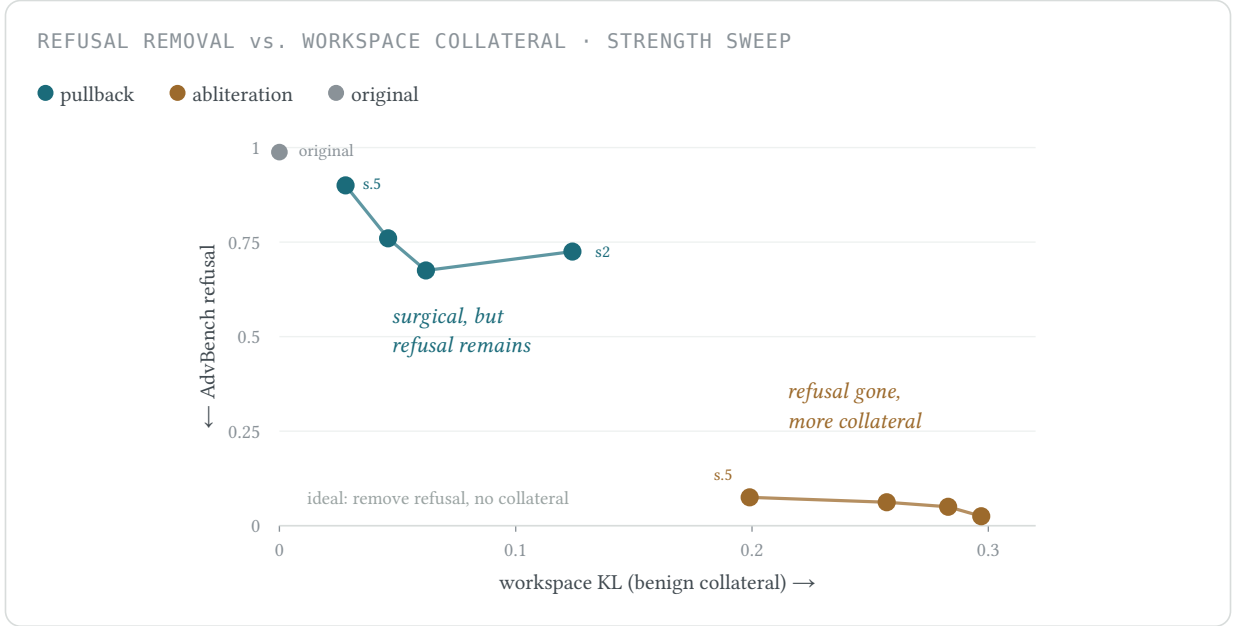


Figure 2: Neither method dominates. Pullback (teal) stays upper-left — low collateral, partial removal; ablation (ochre) reaches full removal but at $4\text{--}6\times$ the workspace collateral.

4 The honest limit

Does capturing *more* of the refusal subspace close the behavioral gap? Yes — but the collateral overtakes ablation. A rank-32 pullback subspace drives AdvBench refusal to 0.10 (matching ablation) while distorting the benign workspace to KL 1.0 — $\sim 4\times$ **ablation's 0.26**.

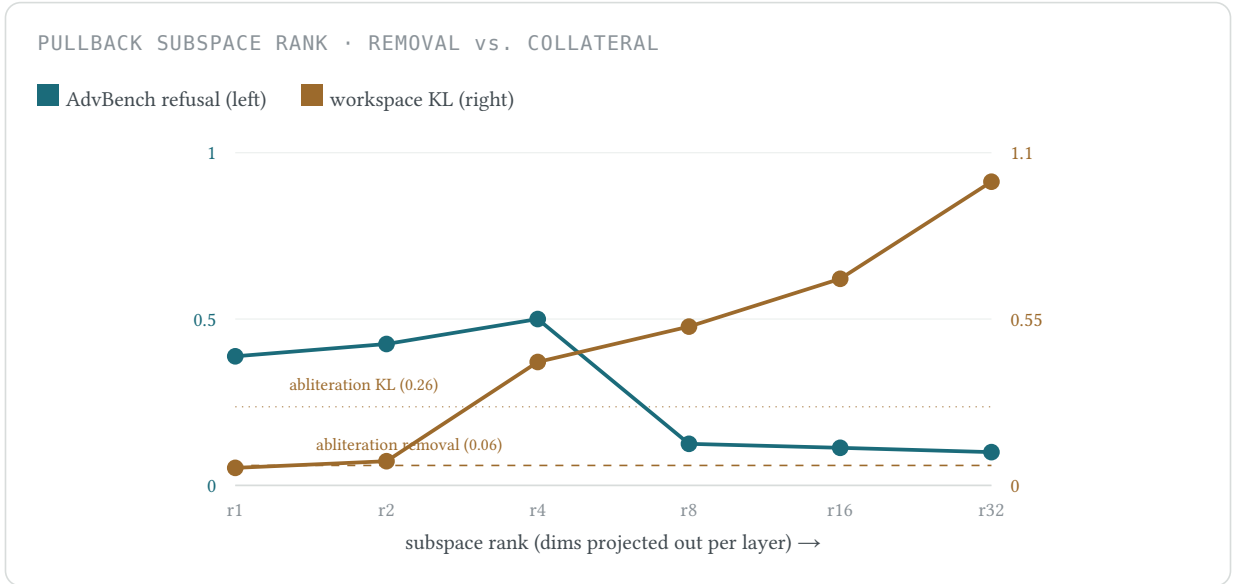


Figure 3: To remove refusal *behavior*, the pullback must project out nearly the whole refusal subspace (rank ~ 32) — and its benign-workspace collateral then crosses above ablation's. The behaviorally decisive part of refusal isn't cleanly separable within the verbalizable workspace.

5 What it means

- **As an instrument, the J-lens is excellent.** It localizes refusal with no contrast set — just the pullback of refusal tokens — reads it before any output, and measures its removal inside the interpretable workspace.
- **As an eraser, it offers what ablation can't:** a graded, low-collateral *partial* dial — remove 30–60% of verbalizable refusal at $KL < 0.08$. But it does not dominate ablation for full removal.

- **Because refusal is only partly workspace-mediated.** Full behavioral removal requires editing directions *outside* the verbalizable workspace — exactly what ablation does, and what it pays for in collateral. The workspace-vs-automatic split, shown causally for refusal.

reproduce · Qwen3.5-4B + pre-fitted Hub lens (neuronpedia/jacobian-lens)
scripts/00_probe_refusal.py · 02_benchmark.py · 03_tradeoff.py · 04_rank_sweep.py
eval on disjoint splits · directions built on a 96/96 fit split